Equação de Euler Lagrange para Campos

$$\mathcal{L} = \frac{1}{2}\rho\,\dot{\phi}^2 - \frac{T}{2}\,\phi_x^2 \quad \text{e} \quad S[\mathcal{L}] = \int_{t_0}^{t}\int_{0}^{\ell} \delta\mathcal{L}\,dx\,dt = 0$$

Note que: $\mathcal{L} = \mathcal{L}(\phi, \dot{\phi}, \phi_x, x, t)$ e $\phi = \phi(x,t)$

$$\phi \to \phi' = \phi + \varepsilon\eta$$

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial \dot{\phi}}\delta\dot{\phi} + \frac{\partial \mathcal{L}}{\partial \phi_x}\delta\phi_x$$

$\to \delta x$ e $\delta t = 0$ pois a única dinâmica vem de ϕ

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial \dot{\phi}}\frac{\partial}{\partial t}(\delta\phi) + \frac{\partial \mathcal{L}}{\partial \phi_x}\frac{\partial}{\partial x}(\delta\phi)$$

$$\delta\mathcal{L} = \left[\frac{\partial \mathcal{L}}{\partial \phi} + \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) + \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial x}\right)\right]\delta\phi = 0$$

$$\frac{\partial \mathcal{L}}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\delta\phi\right) = \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right)\delta\phi + \frac{\partial \mathcal{L}}{\partial \dot{\phi}}\frac{\partial}{\partial t}\delta\phi$$

$$\delta\mathcal{L} = \left[\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \cancelto{0}{\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\delta\phi\right)} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right)\delta\phi + \cancelto{0}{\frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial \phi_x}\cdot\delta\phi\right)} - \frac{\partial}{\partial x}\frac{\partial \mathcal{L}}{\partial \phi_x}\delta\phi\right]$$

termos de fronteira (extremos fixos)

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi}\delta\phi - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right)\delta\phi - \frac{\partial}{\partial x}\frac{\partial \mathcal{L}}{\partial \phi_x}\delta\phi = 0$$

$$\delta \mathcal{L} = \left[\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) - \frac{\partial}{\partial x} \frac{\partial \mathcal{L}}{\partial \phi_x} \right] \delta \phi = 0$$

Portanto:

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) - \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial \phi_x}\right) = 0}$$

P/ $\mathcal{L} = \frac{1}{2} \rho \dot{\phi}^2 - \frac{T}{2} \phi_x^2 \rightarrow \mathcal{L} = \mathcal{L}(\dot{\phi}, \phi_x, t)$

- $\frac{\partial \mathcal{L}}{\partial \phi} = 0$
- $\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \rho \dot{\phi}$ e $\frac{d}{dt}(\rho \dot{\phi}) = \rho \ddot{\phi}$
- $\frac{\partial \mathcal{L}}{\partial \phi_x} = -T \phi_x$ e $\frac{\partial}{\partial x}(T \phi_x) = T \phi_{xx}$ onde $\phi_{xx} = \frac{\partial^2 \phi}{\partial x^2}$

Logo:

$$-\rho \ddot{\phi} + T \phi_{xx} = 0 \Rightarrow \rho \ddot{\phi} - T \phi_{xx} = 0$$

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{T}{\rho} \frac{\partial^2 \phi}{\partial x^2} = 0$$ Eq da onda